Inventory Control with Time-Varying Demand

Characteristics

- Demand varies from period to period
- The demand for each period is exactly known
- Costs may vary from period to period
- Capacity may vary from period to period

Big versus Small Buckets

- Big time-bucket models
  - Items produced/ordered in a period can be used to satisfy the demand for that period
- Small time-bucket models
  - Production/supply leadtimes can take multiple periods

The Lot Sizing Model

Assumptions for the Basic Model

- Demand varies from period to period but is exactly known for each period.
- Demand in each period must be satisfied during the same period (backordering is not allowed).
- There are no limits on how much can be produced or ordered.
- Items produced/ordered in a period are available to satisfy demand during the same period (big bucket model).
- Setup/ordering, production/purchasing, and inventory holding costs can vary period to period.

Objective

*Determine the optimal order quantity (**lot size**) in each period so that the demand in each period is met while the sum of ordering, purchasing, and inventory holding costs are minimized.*

Notation

- t: a period (e.g., day, week, month); $t = 1, \ldots, T$, where T represents the *planning horizon*
- D_t: demand in period t (number of units)
- c_t: unit purchasing/production cost
- A_t: ordering/setup cost associated with placing an order (or initiating production) in period t
- h_t: cost of holding one unit of inventory from period t to period t +1
- Q_t: the size of the order (or lot size) in period t; a decision variable

Example

t	1	2	3	4	5	6	7	8	9	10
D_t	20	50	10	50	50	10	20	40	20	30
c_t	10	10	10	10	10	10	10	10	10	10
A_t	100	100	100	100	100	100	100	100	100	100
h_t	1	1	1	1	1	1	1	1	1	1

The Lot for Lot Solution

t	1	2	3	4	5	6	7	8	9	10
D_t	20	50	10	50	50	10	20	40	20	30
c_t	10	10	10	10	10	10	10	10	10	10
A_t	100	100	100	100	100	100	100	100	100	100
h_t	1	1	1	1	1	1	1	1	1	1

t	1	2	3	4	5	6	7	8	9	10	Total
D_t	20	50	10	50	50	10	20	40	20	30	300
Q_t	20	50	10	50	50	10	20	40	20	30	300
I_t	0	0	0	0	0	0	0	0	0	0	0
Setup cost	100	100	100	100	100	100	100	100	100	100	1000
Holding cost	0	0	0	0	0	0	0	0	0	0	0
Total cost	100	100	100	100	100	100	100	100	100	100	1000

The Fixed Order Quantity Solution

t	1	2	3	4	5	6	7	8	9	10	Total
D_t	20	50	10	50	50	10	20	40	20	30	300
Q_t	100	0	0	100	0	0	100	0	0	0	300
I_t	80	30	20	70	20	10	90	50	30	0	0
Setup cost	100	0	0	100	0	0	100	0	0	0	300
Holding cost	80	30	20	70	20	10	90	50	30	0	400
Total cost	180	30	20	170	20	10	190	50	30	0	700

The Fixed Order Period Solution

t	1	2	3	4	5	6	7	8	9	10	Total
D_t	20	50	10	50	50	10	20	40	20	30	300
Q_t	70	0	60	0	60	0	60	0	50	0	300
I_t	50	0	50	0	10	0	40	0	30	0	180
Setup cost	100	0	100	0	100	0	100	0	100	0	500
Holding cost	50	0	50	0	10	0	40	0	30	0	180
Total cost	150	0	150	0	110	0	140	0	130	0	680

A Mixed Integer Linear Program (MILP) Formulation

$$\textit{Minimize} \qquad z = \sum_{t=1}^{T} c_t Q_t + h_t I_t + A_t Y_t$$

subject to

$$I_t = I_{t-1} + Q_t - D_t, \qquad t = 1,..., T$$

$$Q_t \le Y_t \left(\sum_{t'=t}^{T} D_{t'}\right) \qquad t = 1,..., T$$

$$Y_t \in \{0,1\} \qquad t = 1,..., T$$

$$Q_t, I_t \ge 0 \qquad t = 1,..... T$$

Solution Approach

- Solve as a standard MILP (using for example a *branch and bound* algorithm); several commercial MILP solver software tools are available
- Develop a customized solution that takes advantage of structural properties specific to the problem (e.g., the Wagner-Whitin algorithm)

Property 1

Under an optimal lot-sizing policy either the inventory carried to period t+1 from the previous period will be zero or the production quantity in period t+1 will be zero.

The Basic Idea of the Wagner-Whitin Algorithm

- Using property 1, either Q_t=0 or Q_t=D_t+…+D_k for some k.
- If j_k^* = t = last period of production (or ordering) in a k period problem, then we will produce (or order) exactly D_t+ D_{t+1} …D_k in period j_k^*.
- We can then consider periods 1, … , j_k^*-1 as if they are an independent j_k^*-1 period problem.

The Basic Idea of the Wagner-Whitin Algorithm (Continued…)

- Construct an algorithm where the decision is whether or not to order in a given period. If we order, then the order quantity should be just enough to cover demand until the next period in which we order.

- Solve a series of smaller sub-problems (a one period problem, a two period, …., *N* period problem), where the solution to each sub-problem is used in solving the next subproblem.

Example

t	1	2	3	4	5	6	7	8	9	10
D_t	20	50	10	50	50	10	20	40	20	30
c_t	10	10	10	10	10	10	10	10	10	10
A_t	100	100	100	100	100	100	100	100	100	100
h_t	1	1	1	1	1	1	1	1	1	1

Example

Step 1: Obviously, just satisfy D_1 (note we are neglecting production cost, since it is fixed).

$$z_1^* = A_1 = 100$$

$$j_1^* = 1$$

Step 2: Two choices, either $j_2^* = 1$ or $j_2^* = 2$.

$$z_2^* = \min \begin{cases} A_1 + h_1 D_2, & \text{order in 1} \\ z_1^* + A_2, & \text{order in 2} \end{cases}$$

$$= \min \begin{cases} 100 + 1(50) = 150 \\ 100 + 100 = 200 \end{cases}$$

$$= 150$$

$$j_2^* = 1$$

Example (Continued…)

Step3: Three choices, $j_3^* = 1, 2, 3$.

$$z_3^* = \min \begin{cases} A_1 + h_1 D_2 + (h_1 + h_2) D_3, & \text{order in 1} \\ z_1^* + A_2 + h_2 D_3, & \text{order in 2} \\ z_2^* + A_3, & \text{order in 3} \end{cases}$$

$$= \min \begin{cases} 100 + 1(50) + (1+1)10 = 170 \\ 100 + 100 + (1)10 = 210 \\ 150 + 100 = 250 \end{cases}$$

$$= 170$$

$$j_3^* = 1$$

Example (Continued…)

Step 4: Four choices, $j_4^* = 1, 2, 3, 4$.

$$z_4^* = \min \begin{cases} A_1 + h_1 D_2 + (h_1 + h_2) D_3 + (h_1 + h_2 + h_3) D_4, & \text{order in 1} \\ z_1^* + A_2 + h_2 D_3 + (h_2 + h_3) D_4, & \text{order in 2} \\ z_2^* + A_3 + h_3 D_4, & \text{order in 3} \\ z_3^* + A_4, & \text{order in 4} \end{cases}$$

$$= \min \begin{cases} 100 + 1(50) + (1+1)10 + (1+1+1)50 = 320 \\ 100 + 100 + (1)10 + (1+1)50 = 310 \\ 150 + 100 + (1)50 = 300 \\ 170 + 100 = 270 \end{cases}$$

$$= 270$$

$$j_4^* = 4$$

Property 2

If $j_k^*=t$, then the last period in which ordering/production occurs in an optimal $k+1$ period policy must be in the set t, $t+1,\ldots k+1$.

Property 2 (Continued…)

In the Example:

- We order in period 4 for period 4 of a 4 period problem.
- We would never order in period 3 for period 5 in a 5 period problem.

Example (Continued…)

Step 5: Only two choices, $j_5^* = 4, 5$.

$$Z_5^* = \min \begin{cases} z_3^* + A_4 + h_4 D_5, & \text{order in 4} \\ z_4^* + A_5, & \text{order in 5} \end{cases}$$

$$= \min \begin{cases} 170 + 100 + 1(50) = 320 \\ 270 + 100 \qquad\quad = 370 \end{cases}$$

$$= 320$$

$$j_5^* = 4$$

Step 6: Three choices, $j_6^* = 4, 5, 6$.

And so on.

Example Solution

Last Period with Ordering	Planning Horizon (t)									
	1	2	3	4	5	6	7	8	9	10
1	100	150	170	320						
2		200	210	310						
3			250	300						
4				270	320	340	400	560		
5					370	380	420	540		
6						420	440	520		
7							440	480	520	610
8								500	520	580
9									580	610
10										620
z_t	100	150	170	270	320	340	400	480	520	580
j_t	1	1	1	4	4	4	4	7	7, 8	8

Example Solution (Continued…)

Optimal Policy:

- Order in period 8 for 8, 9, 10 (40 + 20 + 30 = 90 units)
- Order in period 4 for 4, 5, 6, 7 (50 + 50 + 10 + 20 = 130 units)
- Order in period 1 for 1, 2, 3 (20 + 50 + 10 = 80 units)

Note: we order in 7 for an 8 period problem, but this never comes into play in optimal solution.

Heuristics

- Instead of solving the problem optimally, we could use a heuristic (a rule) that leads to reasonably good solutions but not necessarily optimal.
- The advantage of heuristics is ease of implementation and lower computational effort to reach a solution.

Example Heuristics

- Choose a fixed order quantity and order in multiples of this order quantity. Order again when demand in a period cannot be met from available inventory.
- Choose a fixed order period P. Then, every P periods order all the demand for the next P periods.
- Use a greedy heuristic such as the Silver-Meal heuristic.

The Silver-Meal Heuristic

Starting with a period t, order for the next k periods if the resulting average cost per period $z_{t,t+k}$ is smaller than the average cost per period if we ordered only for the next (k-1) periods.